PROGRAMMING IN JAVA

CSA0915

ASSIGNMENT-CO1

CAMPUS SMART-PARKING SYSTEM

TASK 1 – Understand the Problem and Identify the Objects

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1. Inputs and Outputs

Inputs:

- Vehicle registration number
- Vehicle type (Two-wheeler, Car or Electric Vehicle)
- Floor and slot details
- Entry and exit time
- Monthly pass details

Outputs:

- Allocated parking slot
- Unique parking ticket number
- Parking duration
- Total parking fee
- Slot availability status

2. Candidate Classes and Responsibilities

Class	Responsibility
Vehicle	Stores vehicle details
TwoWheeler	Represents two-wheelers
Car	Represents cars
ElectricCar	Represents electric cars
ParkingSlot	Stores slot details and availability
ParkingFacility	Manages floors and parking slots
ParkingTicket	Stores ticket number and entry time
ParkingUser	Stores student, staff and visitor details

3. Methods

- **allocateSlot()** – Allocates a free parking slot.
- **releaseSlot()** – Releases an occupied slot.
- **calculateFee()** – Calculates the parking charge.
- **generateTicket()** – Generates a unique ticket.
- **findVehicle()** – Searches for a vehicle.

4. Business Rules

1. One parking slot can hold only one vehicle.
2. Every ticket must have a unique number.
3. A vehicle must receive a suitable free slot.
4. A released slot becomes available again.
5. Students and staff with valid monthly passes park free.
6. Visitors must pay the applicable parking fee.

5. Assumptions

- The parking facility contains multiple floors.
- Each floor has a fixed number of slots.
- Vehicle registration numbers are unique.
- Parking fees depend on vehicle type and duration.
- Entry and exit times are recorded by the system.

6. Alternative Concept and Complexity Analysis

Instead of using Vehicle objects, vehicles can be stored in parallel arrays:

- regNo[] – Registration numbers
- type[] – Vehicle types
- entryTime[] – Entry times

Operation	**Vehicle Objects**	**Parallel Arrays**
Search vehicle	O(n)	O(n)
Add vehicle	O(1)	O(1)
Add new attribute	Easy	Requires another array

Space Complexity:

Both approaches require O(n) space for n vehicles. However, parallel arrays require separate arrays for every attribute.

Conclusion:

Using objects is better because data and methods are combined in one class. It improves code readability, reusability and maintenance.